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SECS ANALYST REPORT

E/I Balance Sweep Exploration

SECS v1.6 Public Results — Campaign C-005

Campaign	C-005
Author	Analyst
Date	2026-02-28
System	SECS Sovereign
Baseline	
Result	ALL PASS (10/10). Controller converges to a single attractor regardless

PUBLIC RELEASE Empirical Results Reproducible

SECS Sovereign Execution Constraint Substrate

https://github.com/JustNothingJay/SECS_Sovereign

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of initial `targetEntropy`; convergence speed is monotonically related to
distance from observation mean; zero vetoes across entire research envelope.
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System Under Test: SECS Sovereign Execution Constraint Substrate, Phase A **Suite:** 136 suites / 1,730 tests / 0 failures **Campaign Result:** 10/10 pass (5 hypotheses, 10 test cases) **Seeds:** 42, 137, 2026 **Parameters Swept:** `targetEntropy` ∈ [0.05, 0.90], step 0.05, 18 values

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§1 — Question

How does the SECS homeostatic controller behave across a range of `targetEntropy` set-points? Where does it converge quickly, where does it oscillate, and where (if anywhere) does the envelope intervene?

This campaign maps the convergence landscape of a single swept parameter while holding all other parameters at defaults. It produces a topological map of controller behavior, not a verdict.

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§2 — Parameters

Fixed Configuration

Parameter	Value
ConstraintSurface	V1_DEFAULTS
Topology	10 nodes, 12 edges (standard)
Clock	100 ms per cycle (integer stamps)
PRNG	Mulberry32 (per seed)
Observation entropy	uniform [0.1, 1.4]
Observation conductivity	uniform [0.3, 0.8]
Observation mean entropy	0.75
Adaptation cadence	1 (every cycle)

Parameter	Value
Structural plasticity	disabled
Learning	defaults (not swept)
Cycles per run	1,000
Trajectory sample interval	every 5th cycle (200 points per run)
Tail window	last 200 cycles (40 trajectory samples)

Swept Parameter

Parameter	Default	Sweep Range	Step	Values
`homeostatic.targetEntropy`	0.30	[0.05, 0.90]	0.05	18

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§3 — Runs

#	Section	Hypothesis	Configurations	Cycles	Seeds	Runs
1	§A	H18–H20	18 sweep values	1,000	42, 137, 2026	54
2	§B	H21	2 (targets 0.30, 0.70)	1,000	42, 137, 2026	6
3	§C	H22	1 (target 0.30 + control)	1,000	42 (×2), 137	3
4	§D	detail	5 (targets 0.05, 0.30, 0.50, 0.75, 0.90)	1,000	42	5

Total data points: 54 × 200 = 10,800 trajectory samples (landscape) + replay & detail.

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§4 — Observations

§4.1 — Convergence Speed Landscape (H18)

The convergence cycle is defined as the first cycle where

$|$`targetEntropy` $-$ 0.75$|$ < `entropyTolerance`.

targetEntropy	Seed 42	Seed 137	Seed 2026	Mean
0.05	29	23	22	24.7
0.10	29	23	22	24.7
0.15	29	23	22	24.7
0.20	29	23	22	24.7
0.25	29	23	22	24.7
0.30	29	23	22	24.7
0.35	29	22	22	24.3
0.40	29	7	22	19.3
0.45	29	7	13	16.3
0.50	12	6	13	10.3
0.55	2	4	2	2.7
0.60	1	1	2	1.3
0.65	**1**	**1**	**1**	**1.0**
0.70	**1**	**1**	**1**	**1.0**
0.75	**1**	**1**	**1**	**1.0**
0.80	1	2	1	1.3
0.85	1	2	1	1.3
0.90	4	4	1	3.0

Pattern: The convergence landscape is asymmetric around the observation distribution mean (0.75). Configurations at 0.65–0.75 converge in a single cycle — the controller is already within tolerance. Convergence time increases monotonically as the initial target moves below 0.55. For low targets

(0.05–0.35), convergence takes 22–29 cycles — a transient of ~3% of the total run — before the controller reaches the attractor.

The landscape exhibits a plateau below 0.35 (convergence cycle is essentially constant at ~22–29 regardless of initial target), suggesting that the controller's per-cycle correction magnitude is the rate-limiting factor, not the absolute initial distance.

Above 0.75, convergence remains fast (1–4 cycles) even at 0.90, indicating the controller corrects more efficiently when overshooting above the observation mean than when undershooting below it.

§4.2 — Steady-State Attractor

All 54 configurations converge to the same per-seed steady-state attractor:

Seed	Final Mean (targetEntropy, tail 200 cycles)	Final Std
42	0.73375	0.10452
137	0.73625	0.08731
2026	0.79000	0.11522

These values are **identical across all 18 initial targetEntropy settings** within each seed. The initial target determines only the transient duration — the controller erases all memory of the initial condition.

Grand mean across seeds: **0.7533**.
Observation distribution mean: **0.75**.
Deviation from observation mean: **0.0033** (0.44%).

§4.3 — Oscillation Amplitude (H19)

targetEntropy Range	Mean Oscillation Amplitude
Extremes ($\leq$ 0.10 or $\geq$ 0.85)	0.433
Midrange (0.30–0.60)	0.433
All configurations	0.433

Oscillation amplitude is **uniform** across all configurations (0.4 for seeds 42 and 137, 0.5 for seed 2026). The steady-state oscillation is driven entirely

by the stochastic observation sequence, not by the initial target. Since all configurations converge to the same attractor, their steady-state dynamics are identical.

Both extreme and midrange mean oscillation amplitudes are bounded well below 1.0 (assertion threshold), confirming stable controller operation across the entire research envelope.

§4.4 — Correction Density

Seed	Correction Density Range
42	0.935–0.937
137	0.950–0.952
2026	0.936–0.937

The controller emits `homeostatic_correction` events on 93.5–95.2% of cycles, with negligible variation across initial targets (<0.002 within-seed spread). The controller is nearly always active.

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§5 — Stability Profile

Per-Seed Convergence Summary

Metric	Seed 42	Seed 137	Seed 2026
Fastest convergence (cycles)	1 (at 0.60–0.85)	1 (at 0.60–0.75)	1 (at 0.65–0.90)
Slowest convergence (cycles)	29 (at 0.05–0.45)	23 (at 0.05–0.30)	22 (at 0.05–0.40)
Attractor steady-state mean	0.73375	0.73625	0.79000
Attractor steady-state std	0.10452	0.08731	0.11522
Oscillation amplitude	0.4	0.4	0.5
Max correction density	0.937	0.952	0.937

Key Finding: Single Attractor

The homeostatic controller has a **single attractor basin** covering the entire research envelope [0.05, 0.90]. There are no bifurcations, multi-modal attractors, or divergent regimes. Every initial condition converges to the same steady state.

The attractor location is determined by the observation distribution (mean entropy ~0.75), not by the initial parameter setting.

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§6 — Envelope Events

Metric	Value
Total vetoes across 54 runs	**0**
Per-run maximum veto count	0
Per-run veto rate	0.0%
Envelope boundary contacts	0

The entire research envelope [0.05, 0.90] is **unconditionally safe**. No configurations triggered envelope vetoes, warnings, or boundary contacts. The `severeEntropyThreshold` (1.5) is sufficiently distant from the swept range that no interventions occur.

H20 is confirmed: vetoes are zero throughout the research range, as the swept values are well below the constitutional boundary.

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§7 — Determinism Verification

Replay Test (H22)

Metric	Run 1 (seed 42, target 0.30)	Run 2 (seed 42, target 0.30)	Match
Trajectory length	200	200	identical

Metric	Run 1 (seed 42, target 0.30)	Run 2 (seed 42, target 0.30)	Match
Convergence cycle	29	29	identical
Final mean	0.73375	0.73375	identical
Final std	0.10452	0.10452	identical
Oscillation amplitude	0.4	0.4	identical
Veto count	0	0	identical
Correction density	0.935	0.935	identical
Full JSON comparison	—	—	**byte-identical**

Control (Seed Divergence)

Metric	Seed 42	Seed 137	Match
Trajectory JSON	(unique)	(unique)	**different**

Interpretation: determinism is per-seed, not substrate-wide. Different PRNG seeds produce different observation sequences, yielding different transient paths to the same attractor.

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§8 — Cross-Seed Consistency (H21)

At targetEntropy = 0.30

Metric	Seed 42	Seed 137	Seed 2026
Convergence cycle	29	23	22
Final mean	0.73375	0.73625	0.79000

Metric	Seed 42	Seed 137	Seed 2026
Final std	0.10452	0.08731	0.11522
Correction density	0.935	0.951	0.936
Mean spread	—	—	**0.056**

At targetEntropy = 0.70

Metric	Seed 42	Seed 137	Seed 2026
Convergence cycle	1	1	1
Final mean	0.73375	0.73625	0.79000
Mean spread	—	—	**0.056**

Mean spread 0.056 < threshold 0.15 at both probed targets. Convergence patterns are qualitatively similar across seeds: same attractor, same steady-state dynamics, seed-specific differences only in transient timing.

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§9 — Detailed Trajectories (§D)

Five key configurations were recorded at full resolution (seed 42):

targetEntropy	Conv Cycle	Final Mean	Final Std	Osc Amp	Veto	Corr Density
0.05	29	0.73375	0.10452	0.4	0	0.935
0.30	29	0.73375	0.10452	0.4	0	0.935
0.50	12	0.73375	0.10452	0.4	0	0.935
0.75	1	0.73375	0.10452	0.4	0	0.937
0.90	4	0.73375	0.10452	0.4	0	0.937

All five configurations reach the same steady state. Only the convergence cycle differs — confirming that `targetEntropy = 0.75` provides the fastest initial convergence, while low values (0.05, 0.30) require ~29 cycles to traverse the distance to the attractor.

Trajectory data (200 points per run, 1,000 points total) is emitted in full in the test output log for archival.

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§10 — Disposition

Per-Hypothesis Summary

ID	Hypothesis	Result	Evidence
H18	Convergence speed varies with target proximity	**PASS**	Convergence cycle 1 at 0.65–0.75 vs 22–29 at 0.05–0.35
H19	Oscillation amplitude bounded across configs	**PASS**	Uniform 0.4–0.5, both extreme and midrange < 1.0
H20	Zero vetoes within research envelope	**PASS**	0 vetoes across all 54 runs
H21	Cross-seed qualitative consistency	**PASS**	Mean spread 0.056 < 0.15 at both probed targets
H22	Deterministic replay	**PASS**	Byte-identical trajectories; control confirms seed divergence

Campaign Result: 10/10 PASS

Interpretation Notes (Computational Only)

1. **Single attractor:** The controller erases initial conditions within ~30 cycles at worst. The steady state is fully determined by the observation distribution, not the initial parameter.
2. **Asymmetric transient:** The convergence landscape is asymmetric — the controller corrects more rapidly from above the attractor (0.90 → 4 cycles) than from below (0.05 → 29 cycles).
3. **Plateau regime:** Below targetEntropy 0.35, convergence time is essentially constant (~24.7 cycles), suggesting a constant-rate correction mechanism rather than proportional control.

4. **Research envelope safety:** The entire [0.05, 0.90] range is unconditionally safe (zero vetoes). Future campaigns can explore the full range without envelope concerns.
5. **Seed independence:** The attractor location and oscillation amplitude are consistent across seeds. Per-seed variation is limited to the specific value of the attractor mean ($\pm$0.056), determined by the PRNG-generated observation sequence.

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§11 — Constitutional Compliance

Constraint	Status
Zero substrate modifications	Verified — no changes to `src/`
No clinical claims	Verified — computational language only
`Type: COMPUTATIONAL EXPLORATION` header	Present
Parameters within research envelope	All 18 values in [0.05, 0.90]
Deterministic replay confirmed	H22 PASS
Report uses permitted fields only	Verified against `research-report-schema.md`
No prohibited fields	No diagnosis, clinical_recommendation, biological_claim, treatment_suggestion, patient_identifier, medical_interpretation, or neurotypical_comparison

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First research campaign complete. One axis swept. Landscape mapped. The controller has a single attractor — initial conditions are forgotten.